WILDAN IBRANSYAH

Jember, East Java
+62 823 1167 8297 | wildanibrans@gmail.com | linkedin.com/in/wildanibransyah

SUMMARY

Fresh graduate in D-IV Mechatronics Engineering Technology from Politeknik Negeri Jember with a focus on industrial automation and hardware integration. Experienced in control panel wiring, HMI design, and computer vision-based object detection through internship and academic projects. Proven ability to deliver measurable results, including a 98.7% precision object detection model for real-world agricultural application. Open to opportunities in industrial automation.

WORK EXPERIENCE

PT. Mokko Otomasi Indonesia, Gresik — Aug – Nov 2025
Internship
- Diagnosed and resolved wiring faults on the control panel of a shrink tunnel machine, ensuring proper electrical connection and system integrity.
- Conducted field surveys and installed 2 water level sensors (hulu and hilir) at Bendungan Mlirip, Mojokerto and Bendungan Jagir, Surabaya under PERUM JASA TIRTA I for remote monitoring systems.
- Redesigned HMI interface layout at PT. Warnatama Cemerlang, improving operator monitoring efficiency and reducing manual control errors.

CV. Lab Persada, Lumajang — Jan – Apr 2019
Internship
- Diagnosed and repaired approximately 15 laptops (hardware & software), restoring device functionality through OS reinstallation and component troubleshooting.
- Completed field services including CCTV installation and hardware assembly based on client requirements.

EDUCATION

Politeknik Negeri Jember — 2022 – 2026
D-IV Mechatronics Engineering Technology - GPA: 3.72/ 4.00

Relevant Coursework: Mikrokontroller, PLC Dasar & Lanjut, SCADA dan HMI, Sensor dan Transducer, IoT, Perancangan Sistem Mekatronik

CERTIFICATIONS

- Engineer Mekatronika — Badan Nasional Sertifikasi Profesi (BNSP)
- Computer Network Engineer (Teknik Komputer Jaringan) — Badan Nasional Sertifikasi Profesi (BNSP)

PROJECTS

Final Project — 2026
YOLOv11n Implementation on Raspberry Pi for Cacao Fruit Physical Condition Detection
- Collected an initial dataset of 530 cacao fruit images and expanded it to 1,643 images through data augmentation using Roboflow to improve model generalization.
- Trained and evaluated a YOLOv11n object detection model deployed on Raspberry Pi, achieving 98.7% Precision, 99% Recall, and 99.4% mAP50 across 10 real-time test scenarios.
- Addressed challenges in real-time inference optimization on edge devices to balance detection accuracy with limited hardware computational resources.

IoT-Based Egg Incubator Prototype — 2024
- Engineered an IoT-based monitoring system integrating ESP32 microcontroller and sensors for real-time temperature and humidity control.

- Designed the control logic to maintain stable environmental conditions to support automated incubation, integrating sensor feedback for consistent monitoring.

ORGANIZATIONAL EXPERIENCE

SENO (Sinergi Energi dan Otomotif) — PDD Division, PLC National Competition 2024

- Operated presentation slides during the opening ceremony and technical rules briefing session, ensuring smooth delivery of information to competition participants.

SENO (Sinergi Energi dan Otomotif) — Consumption Division, Energy Seminar 2025

- Coordinated consumption logistics for a campus-level energy seminar, ensuring smooth event operations for attendees and speakers.

SOFT SKILLS & TECHNICAL SKILLS

Soft Skills

- Communication
- Teamwork
- Adaptability
- Problem Solving
- Analytical Thinking
- Eager to Learn

Embedded Systems

- Raspberry Pi
- ESP32
- Arduino
- Sensor Integration
- C/C++
- Python

Industrial Automation

- Basic PLC Programming
- HMI Design
- Control System Installation
- Panel Wiring & Assembly
- Troubleshooting

Software

- Microsoft Office
- Visual Studio
- Canva
- Factory I/O